1 Conic Sections

When a double cone is sliced by a plane, the cross section is formed by the plane and the cone is called a **Conic Section**. The four main conic sections are the circle, the parabola, the ellipse, and the hyperbola.

1.1 Circle

A **circle** is a set of point which are equidistant from one point. The one point being called the **center** and the distance of the line is the **radius**. The standard form equation with the center at (0,0) is: $x^2 + y^2 = r^2$

1.2 Parabola

A **parabola** is the set of points in a plane that are the same distance from a given point and a given line in that plane.

The given point is the **focus** and the line is called the **directrix**.

The midpoint of the perpendicular segment from the focus to the directrix is called the **vertex** of the parabola.

The line which passes through the vertex and focus is called the **axis of symmetry**.

1.3 Ellipse

An **Ellipse** is the set of points in a plane such that the sum of the distances from two given points in that plane stays constant.

The two points are each called a **focus point** – plural *foci*.

The midpoint of the segment joining the foci is called the **center** of the ellipse.

Ellips have two axes of symmetry:
Longer one = **Major Axis**, Shorter one = **Minor Axis**.
The **major intercepts** are where the 2 points intercept with the major axis.
The **minor intercepts** are where the 2 points intercept with the minor axis.

1.4 Hyperbola

A **hyperbola** is the set of all points in a plane such that the absolute value of the difference of the distances between two given points stays constant.

Two given points are the **foci** of the hyperbola.

Hyperbolas looks like two opposing 'U-shaped' curves.

Hyperbolas have two axes of symmetry:
The one going through the hyperbola's foci is the **transverse axis.**
The one going through the center and is perpendicular to the transverse is the **conjugate axis**.